The Gwangyang earthquake sequence, 2012–2026: study plan

Detection, relocation, template matching and magnitudes with time-dependent sensor orientation

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Summary. Beneath Gwangyang, on the south coast of Korea, the Korea Meteorological Administration (KMA) listed 6 earthquakes in July–August 2021 and 5 in January–February 2026, of local magnitude M_L 1.1–2.2. An earlier precise relocation places all 11 within about 141 m of each other at 13.9 km depth. This study rebuilds the earthquake history of that spot from 20 December 2012, when the nearest broadband station (KS.GWYB) began recording, to 6 October 2026. It uses the same stages as the 2025 Imsil study: deep-learning detection, double-difference relocation, local magnitudes, template matching over the whole period, and joint relocation of the new detections. What is new is how the horizontal sensors are handled. Their orientation changes over time: GWYB alone sits at four different angles in this period, and none of them is recorded in the station metadata. We therefore apply the current Haenam rules: one sensor per station, one dated orientation table, S-wave timing measured on the transverse component, and templates turned into each day's orientation. The plan answers five questions: whether the spot was active before, between and after the two episodes; whether 2026 re-ruptured the 2021 patch; the geometry of the patch; how events spread within each episode; and the magnitudes and completeness under a network that grew from 8 stations within 60 km in 2013 to 29 in 2026. The data, the sensor orientations and the smoke test are done, and the detection settings are frozen. Production detection waits for the graphics processor, which the higher-priority Haenam work holds until about 19 October. A pilot on parts of the two episode years found 21 events near the patch that KMA did not list. 19 of them have the S−P intervals of the KMA events and correlate with their waveforms, so they come from the same patch (Section 7). Among them are 3 small events that preceded the first KMA-listed event of 2021 by 1.5–2.1 days.

1 Questions

The two episodes are small, but they occurred on the same spot five years apart, in a stable continental interior where repeated activity at one place is rarely documented. KMA also lists 14 other earthquakes within 10 km of the spot since December 2012 (Figure 1a). Three of them lie within 5 km: 22 April 2019 (ML 1.0, 3.5 km); 26 June 2023 (ML 1.5, 3.5 km); 26 May 2026 (ML 1.6, 0.8 km). KMA locations are uncertain by kilometers, so whether any of them belong to the same fault patch is unknown. The study asks:

Q1. **History.** Was the patch active before 2021, between the episodes and after February 2026? This includes the 14 other KMA events and events too small for the KMA catalog.

Q2. **Repeat or neighbor.** Did the 2026 episode re-rupture the 2021 patch, or a part next to it?

Q3. **Geometry.** The orientation and size of the active fault plane, with uncertainties.

Q4. **Migration.** How events spread in space and time within each episode.

Q5. **Size and completeness.** Local magnitudes on one consistent scale, and the smallest magnitude reliably detected in each part of 2012–2026 as the network changed.

2 The sequence and earlier work

The 11 KMA events occurred from 20 July to 28 August 2021 and from 1 January to 4 February 2026 (UTC). The largest, M_L 2.2, occurred at 2021-08-27 22:03:22 UTC (2021-08-28 07:03 Korean time). An earlier relocation of these 11 events used deep-learning picks and double-difference relocation with waveform cross-correlation (Waldhauser and Ellsworth, 2000), with 1,000 bootstrap replicas for the uncertainties. It places them within 141 m horizontally and 45 m in depth around 13.9 km, with median 95 % half-widths of 3.2, 2.2 and 5.5 m in east, north and depth (Figure 1b, c). The 2026 events lie inside the 2021 cloud. That relocation is the independent check for this study (Section 8). It does not answer Q1–Q5, because it holds only the KMA-listed events.

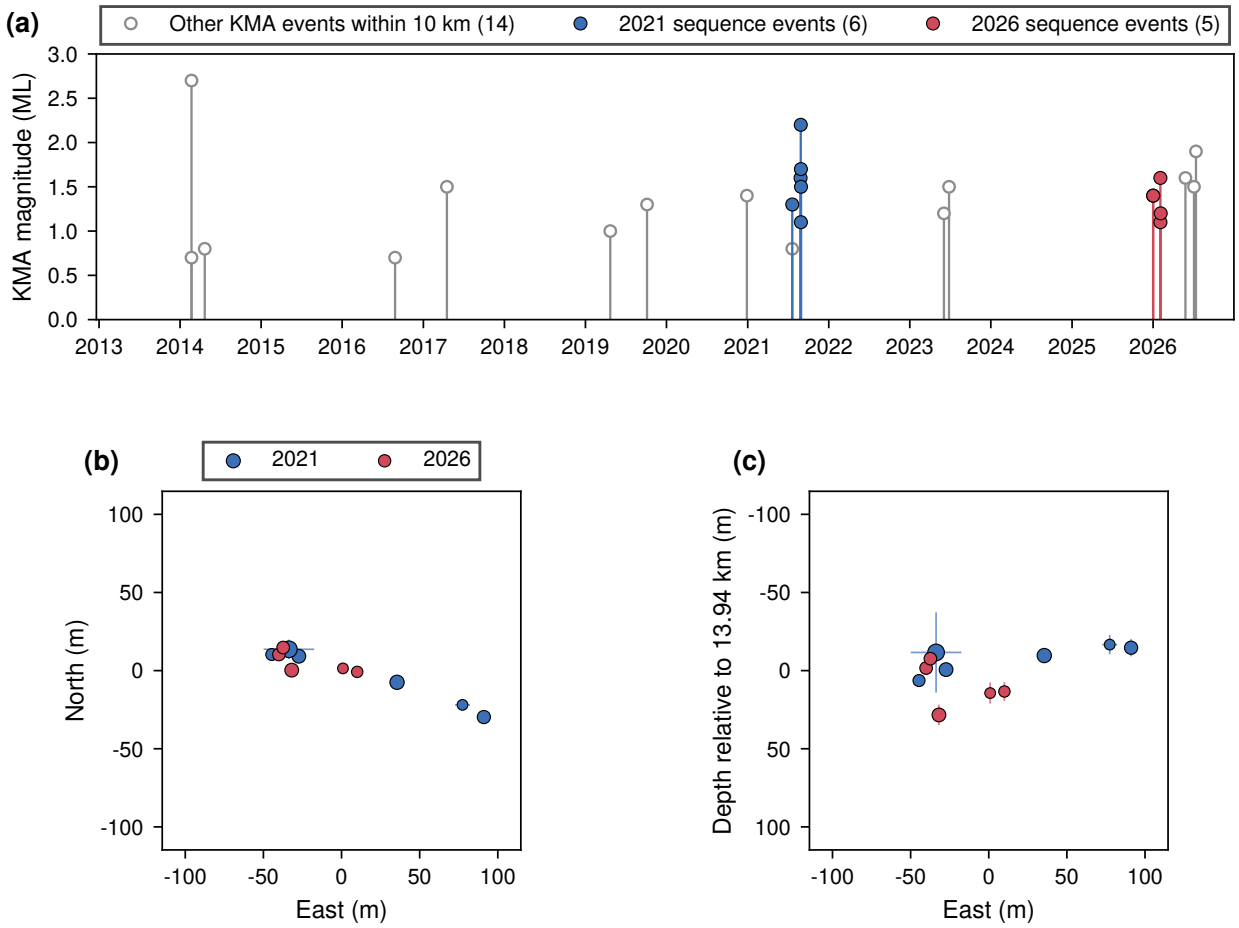


Figure 1. The sequence. (a) KMA-listed earthquakes within 10 km of the cluster centroid (35.0000° N, 127.7286° E) since 20 December 2012 (NECIS domestic catalog, UTC): the 6 events of 2021 and the 5 of 2026 in color, the 14 others open (2014 (3), 2016 (1), 2017 (1), 2019 (2), 2020 (1), 2021 (1), 2023 (2), 2026 (3)). (b) Map view and (c) east–depth section of the earlier relocation of the 11 sequence events; symbol size grows with magnitude, bars are bootstrap 95 % half-widths.

3 Approach

The stage structure, notebooks, decision log and report pattern are those of the 2025 Imsil study. The code is copied from the current Haenam tree (the 2026 Haenam renewal), not from the Imsil tree. The Haenam tree holds the orientation rules of 11–14 October 2026, which the Imsil port predates. Of the copied files, 95 are byte-identical to the Haenam files and 12 carry marked edits: region values, paths, and the inventory and link rules. Detection uses the frozen Haenam settings and network weights, which were validated on the 2024 Buan sequence (Seo et al., 2025).

Why sensor orientation matters. A seismometer's two horizontal components are meant to point north and east. In practice the north component can point elsewhere, and the angle changes when an instrument is installed, serviced or replaced. This angle is called the *misorientation*. Two steps of the study depend on it:

- **S-wave timing.** For relocation, S arrival-time differences between events come from cross-correlation of the horizontal records. We measure them on the *transverse* component, the horizontal motion perpendicular to the source–station direction, where S is strongest and P weakest. Computing the transverse component needs the angle valid on the day of each event. A wrong angle mixes components and can bias the measured time.
- **Template matching.** A template recorded in 2021 is compared with the records of every day from 2012 to 2026. If the sensor was turned between the two days, its north–east records no longer match.

The rules, decided for Haenam and applied here unchanged (decision P3), are:

1. one sensor per station for the whole period, chosen to have the fewest orientation changes (ties: broadband, then short period, then accelerometer);
2. one dated orientation table, used by every stage;
3. S differential times on the transverse component, P on the vertical;
4. template north–east windows turned into each scanned day's orientation, with the continuous data never rotated;
5. local magnitudes from every station with a valid instrument response, without station corrections.

4 Data

4.1 Stations

The network within 100 km of the cluster holds 110 station codes: 102 of KMA (network KS) and 8 of the Korea Institute of Geoscience and Mineral Resources (KIGAM, network KG) (Figure 2). Five temporary codes of 2023 (DAYAT, GEJAT, GWYBT, NAWBT, SUCAT) are read as their own sites. The nearest station is GWYB (7.1 km). The chosen sensors are 14 broadband (HH), 14 short-period (EL) and 82 accelerometers (HG). The sensor rule above has one addition for a 14-year window (decision G-D1b). A sensor whose record is more than a year shorter than the station's longest is not eligible. Without it, stations such as BOSB, NAWB, HAMB and KOHB would be read on accelerometers that start only in 2016, and they would drop out of 2012–2015. Where a station moved within 1 km and changed its code, the older code is used until the day before the newer one starts (G-D10). One code, KS.KWJ, is excluded (G-D20): its files on MARA are numbered in sequence rather than by day of year and sampled at 80 Hz, so linked by name they would put April data on 1 January.

The network grew strongly over time (Figure 3a). Within 60 km, 8 stations recorded in 2013, 16 in 2016, 24 in 2021 and 29 in 2026. Within 50 km the counts are 4, 11, 15 and 19. Five early stations recorded only on 20-sample-per-second short-period channels until March 2015, the closest being YSU at 16.6 km. These channels have no instrument response in any metadata source and cannot carry the 5–20 Hz band of the cross-correlation. They are not used for now (G-D12, open).

4.2 Continuous data

Three sources are used, each for one purpose:

- **MARA**, the laboratory's copy of the KMA and KIGAM archives, provides almost everything: 237,867 station-days of the chosen sensors, 6.9 TB, mirrored onto the analysis machine. A verification against MARA finds only files of 8 October 2026, after the window, missing.
- **KIGAM's data service (FDSN)** provides the KG stations after 31 October 2024, where MARA ends: 3,543 station-days, fetched as 10,629 files with 0 errors (G-D14). Without it, JRB, GSU and GRE, all within 40 km, would be missing from the whole 2026 episode.
- **NECIS**, KMA's public data portal, fills only the holes in MARA (G-D13). KMA's own archive, of which MARA is a copy, lacks some channels on some days. On 2021-08-27, the day of the M_L 2.2, the vertical component is missing at seven nearby stations while the horizontals are present. NECIS sometimes still holds such files.

Every stage reads one tree of links to the chosen sensors' day files: 711,715 links, which pass a consistency check (no dangling, misplaced or missing link).

4.3 Completeness of the episode data

A check of every station and day found 2026 complete. In 2021 it found 14 station-days with only the vertical missing, all inside the episode, and 10 whole station-days missing. 8 of 11 requests to NECIS recovered files (18 day files), among them GWYB's vertical on 28 August and BAU's whole day on 27 August. Over

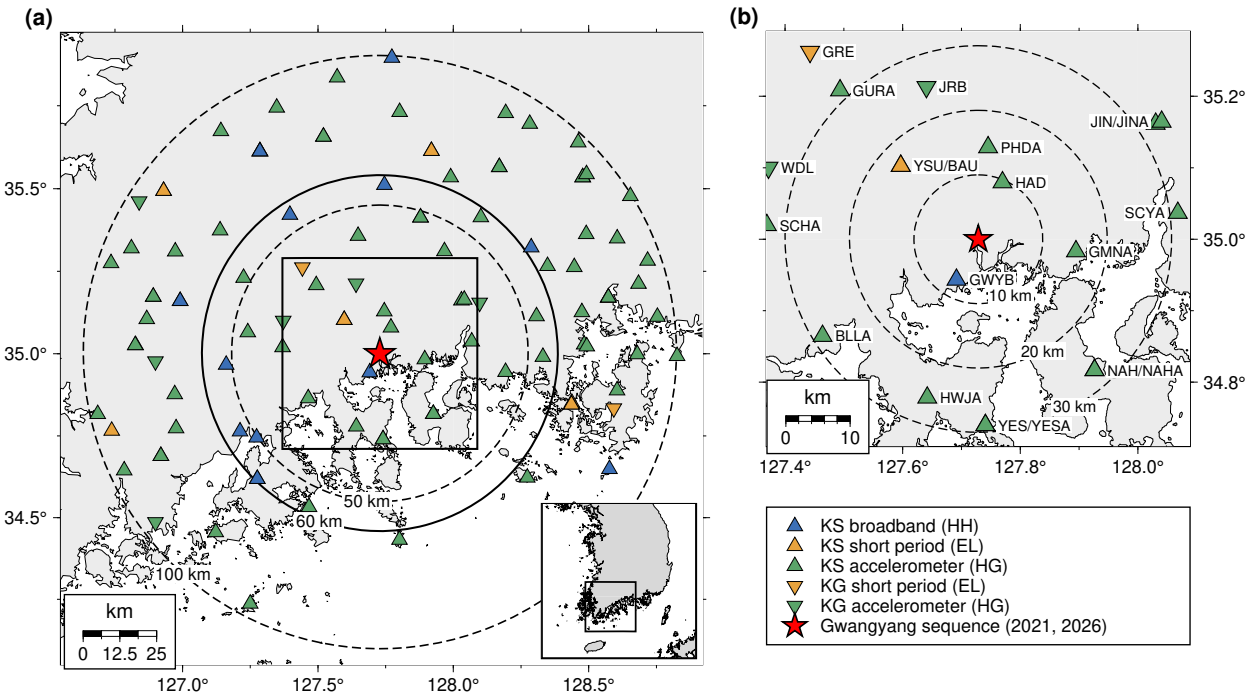


Figure 2. Stations. (a) The 110 station codes within 100 km of the cluster (star), colored by the chosen sensor; triangles KS, inverted triangles KG. The solid circle (60 km) bounds the template-matching stations. Inset: location in Korea. (b) The box of panel (a); codes that replaced each other at one site are joined by a slash.

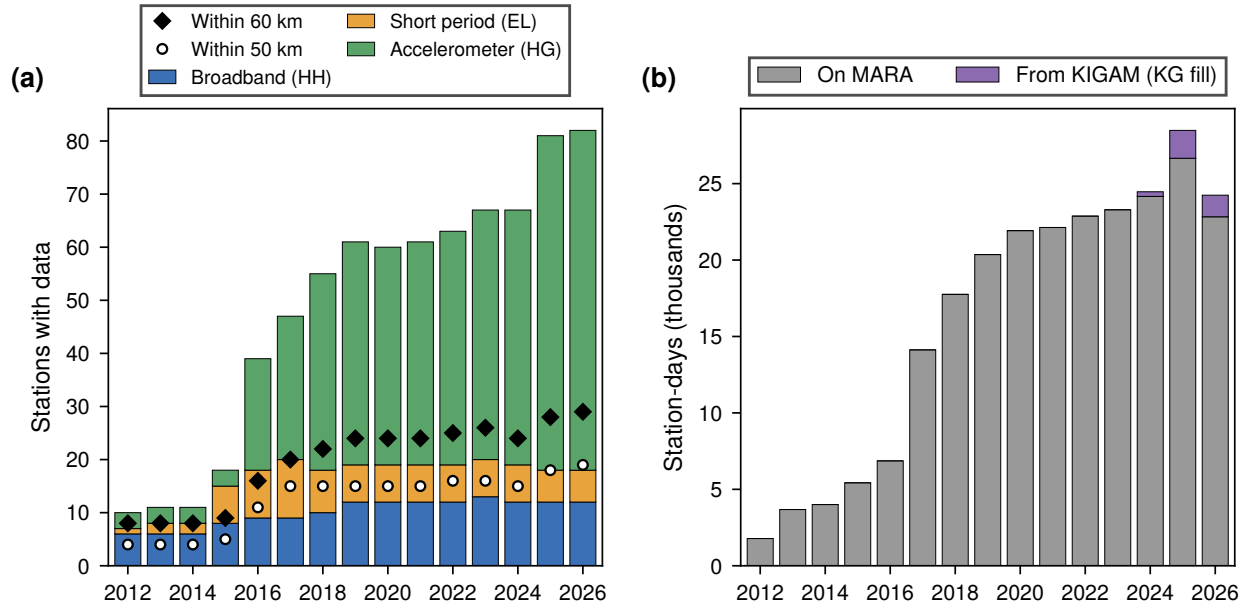


Figure 3. Stations and data per year. (a) Stations within 100 km with data from their chosen sensor, by sensor type; diamonds and circles count those within 60 and 50 km. (b) Station-days of the chosen sensors: on the MARA archive, and fetched from KIGAM (KG stations after October 2024). Whole calendar years are counted; the study starts on 20 December 2012.

July–September 2021, 99.8 % of the station-days of the 61 stations are now complete (Figure 4). 18 station-days are complete only thanks to NECIS, and 4 still lack the vertical. Three of those four belong to BOSB and GMNA, two stations that have since closed and that NECIS no longer serves. Over the whole window, only 8 station-days in 2012–2026 lack a component.

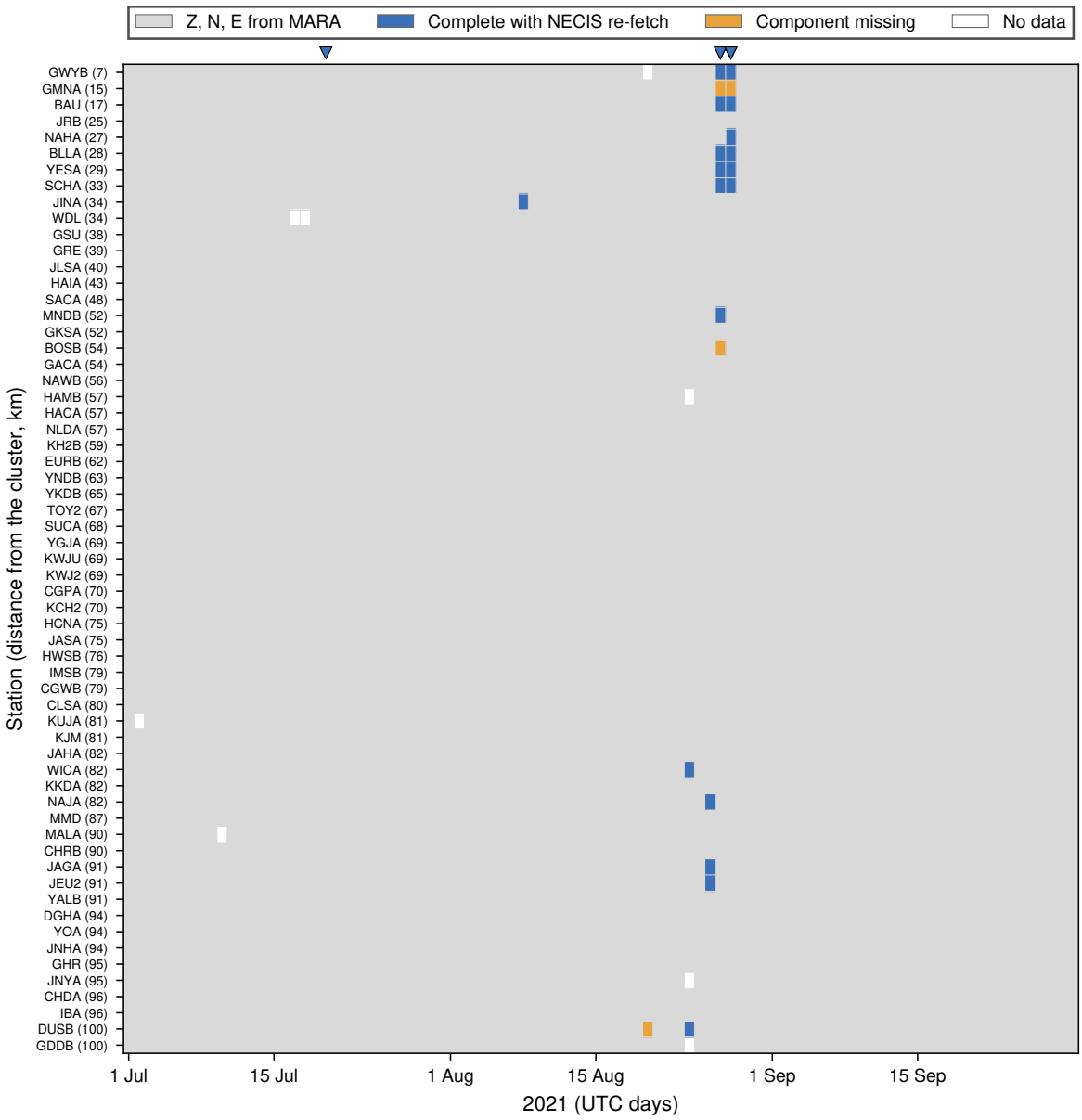


Figure 4. Data of the 2021 episode, day by day, for the 61 stations that recorded between 1 July and 30 September 2021, sorted by distance from the cluster (km in brackets). Gray: all three components from MARA; blue: complete with components re-fetched from NECIS; orange: a component missing; white: no data. Triangles: days of the KMA-listed 2021 events.

5 Sensor orientation

Method. The misorientation is measured from the direction of P-wave motion of distant large earthquakes (teleseisms of magnitude 5.5 or more). A P wave moves the ground along its direction of travel, so on a correctly oriented sensor the horizontal motion points toward the source. The difference between the observed and the expected direction is the misorientation (Seo et al., 2022). The national study of all Korean stations ('33' below) applied this method to every KS and KG station up to August–September 2026. It found the periods of constant angle, wiring faults, silent channels and gain errors. Each change is dated by a metadata epoch inside its bracket where one exists. This study takes those tables. It measures, with the same code and settings, what they lack: the Gwangyang sensors never measured, the years before the national study's data begin, and every station to the end of the window.

Check of the local measurement. The national study cut its windows from KMA's waveform server; this study cuts them from the downloaded day files. On the 950 teleseisms that both measured and both accepted, the two angles differ by a median of 0.00°, a 95th percentile of 0.10° and at most 0.30° (Figure 5c).

Result. The final run on the complete data measured 22,078 station–teleseism windows. It gives 147 periods of constant angle for 110 sensors (144 valid). GWYB, 7 km from the cluster, has four periods (Figure 5a):

- −125.5° from 2013-01-10 to 2016-09-23;
- +94.6° from 2016-10-15 to 2017-07-26;
- −95.2° from 2017-08-11 to 2023-07-02;
- +45.6° from 2023-07-03 to the end.

These rest on 1,404 accepted measurements. The two episodes therefore fall in different periods: 2021 in the third and 2026 in the fourth. Of the 39 chosen sensors within 60 km, 9 changed orientation at least once (Figure 5b). Overall, 104 chosen sensors have at least one measured period. 6 have none (CHR, HAC, JAH, JAS, SAC, YES), all short-lived accelerometers of 2016–2018. A sensor that stays unmeasured is treated as unturned and flagged (G-D2). On days inside a change bracket no transverse component is formed and the template's S windows get zero weight.

6 Processing stages

Figure 6 and Table 1 give the ten stages. Every script runs as a dry run unless told to write, and each stage ends with a numbered notebook. That notebook holds its figures, its pass or fail lines and a computed summary. Detection, cross-correlation and template matching never share the graphics processor, neither with each other nor with other projects' jobs on the same machine. Before production detection, a *pilot* on parts of the two episode years exercised the detection code (Section 7). It runs in a separate folder, and no production stage reads it.

Table 1. Stages, settings and checks. The detection and relocation settings are the frozen Haenam values.

Stage	Method and settings	Check
G0 Setup	Copy of the Haenam tree with checksums; project registry of paths; decision log	port check: 95 identical, 12 marked edits
G1 Stations, data	Station inventory; MARA mirror; KIGAM fill; NECIS gap filling; one link tree of the chosen sensors	every chosen station-day audited for missing components
G2 Orientation	Teleseismic P polarization (Seo et al., 2022); national tables plus local measurements; one dated table; final sensor choice	local versus national angles; sensor rule applied to every multi-sensor station
G3 Smoke test	Full detection chain on 2021-08-27/28, 2026-01-01 and 2013-07-19	passed: 7 of 7 KMA events within 1.23 s; M_L 2.2 0.12 km from the earlier relocation; picks within 0.019 s (P) and 0.020 s (S) of manual picks (median, 458 picks)
G4 Detection	PhaseNet+, the multi-task successor of PhaseNet (Zhu and Beroza, 2019), at 100 Hz, probability ≥ 0.2; association with PyOcto (Münchmeyer, 2024), ≥ 6 picks (≥ 3 P, ≥ 3 S) in 34.30–35.70° N, 126.93–128.53° E; HYPOINVERSE (Klein, 2002) in the model of Kim and Kim (1983)	all 14 KMA events within 5 km and the other KMA events within 10 km; regional KMA recovery; activity per year
G5 Relocation	Events within 5 km of the centroid, depth ≥ 10 km, gap $< 180°$, > 7 phases, RMS < 1 s, errors < 5 km; double difference (Waldhauser and Ellsworth, 2000) with catalog and cross-correlation times (5–20 Hz, ± 0.5 s, 1,000 Hz interpolation, CC ≥ 0.7, ≥ 4 per pair), P on the vertical and S on the transverse; 1,000 bootstrap replicas; *reference set* = events with 95 % half-widths ≤ 25 m in east, north and depth; plane fitted to the reference events within 1 km of the M_L 2.2	relative positions against the earlier relocation after one rigid shift; S on the transverse against north–east at stations that never turned; how links survive across instrument changes
G6 Magnitudes	Sheen et al. (2018) scale; every station with a valid response on the event day (merged metadata of KMA, KIGAM and NECIS); signal-to-noise ≥ 2; ≥ 3 picked stations; no station corrections	our M_L against KMA M_L for the listed events
G7 Template matching	Templates = reference set; stations within 60 km, one sensor each; 5–20 Hz; 4 s windows from 1 s before P (vertical) and S (north, east); template north–east turned into each day's orientation; per-channel signal-to-noise ≥ 2; detection ≥ 10 times the median absolute deviation of the network correlation, per station configuration, with ≥ 4 stations (Beaucé et al., 2018)	self-detection of every template; GWYB templates detected in all four orientation periods; the Haenam threshold (12 times plus a correlation floor of 0.2) compared on the episode days; evidence notebooks for noise bursts and single-template families
G8 Joint relocation	New detections inherit their template's picks (only to place the windows) and enter by cross-correlation times alone, with the references; S on the transverse; G5 settings, bootstrap and 25 m rule	reference events move little; detection pass rate versus correlation
G9 Analysis, report	Activity history by episode; repeat or neighbor; plane; migration; magnitudes and completeness per network era; report with the same fact system	answers to Q1–Q5 with uncertainties

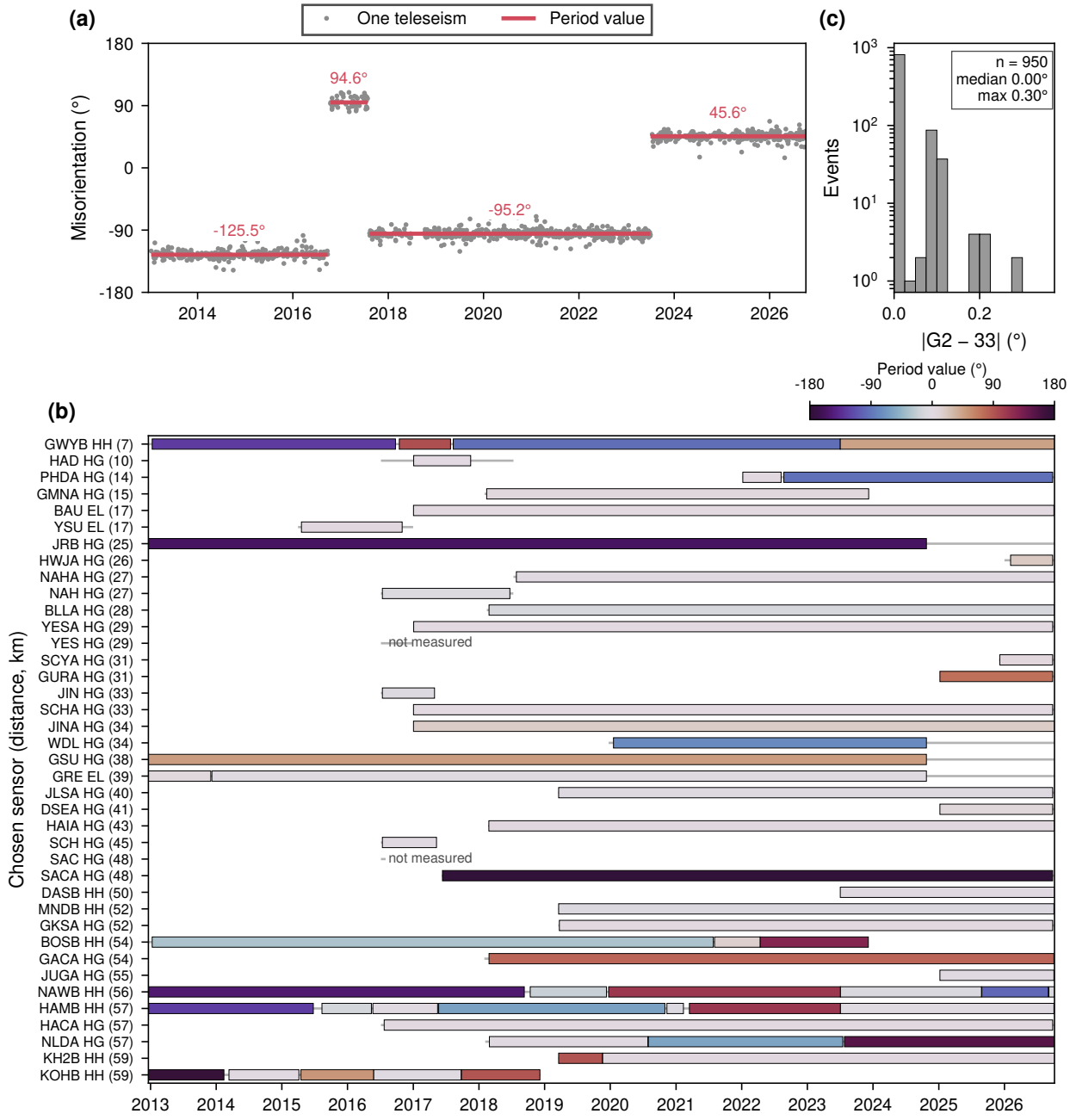


Figure 5. Sensor orientation. (a) KS.GWYB broadband: the misorientation measured on each accepted teleseism (gray) and the four period values of the orientation table (red). (b) Orientation periods of the chosen sensors within 60 km, colored by angle; thin gray lines are the data spans, and a gap between colored bars is an unresolved change bracket. (c) The same teleseisms measured from the local day files (this study, G2) and from the national study's server windows (33): absolute difference of the accepted angles.

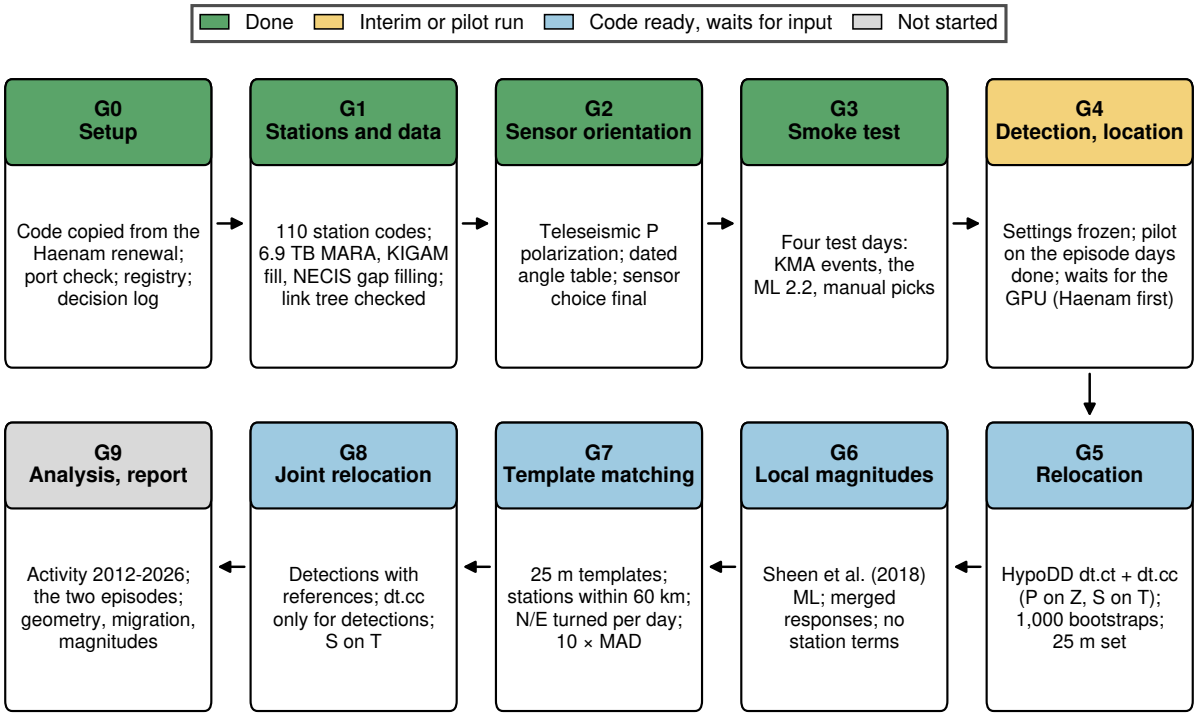


Figure 6. Processing stages and their status as of 2026-10-16 16:40 KST.

7 Pilot results on the episode days

The pilot ran the frozen detection chain on 2021 days 1–64 and 184–245 (1 January to 5 March and 3 July to 2 September) and on 2026 days 1–51 and 140–190 (1 January to 20 February and 20 May to 9 July), with the station table from before the freeze (no KG stations in 2026). The relocation selection of G5 keeps 33 events: 11 on the KMA sequence list, 1 other KMA-listed event (26 May 2026, 15:45 UTC, ML 1.6) and 21 not in the KMA catalog. Appendix B lists them all. These are single-event HYPOINVERSE locations; the cross-correlation relocation comes with the production run (G5). KMA lists 14 events within 5 km of the patch since December 2012, three more than the sequence list. The production detection is checked against all of them (decision R5).

Locations. The KMA-listed events locate with a median of 84 phases, a gap of 35°, ERH 0.2 km and ERZ 0.3 km; the new events with 16 phases, a gap of 86°, ERH 0.5 km and ERZ 0.7 km (Figure 7, Figure 8). The median largest principal 1σ error is 0.29 km for the KMA-listed events and 0.73 km for the new ones. The 9 new events with 12 or fewer phases or a vertical error above 1 km come out shallower and off the patch, with error ellipses elongated toward the surface. A double-difference relocation from the picks alone already draws 31 of the events together to standard deviations of 110 m east, 188 m north and 214 m in depth, against 749 m, 366 m and 1165 m east, north and in depth for HYPOINVERSE.

Same patch? Two tests do not depend on the locations (Figure 9, Figure 10).

- **S−P intervals.** At a given station, events at the same place have the same S−P interval. 20 of the 21 new events match the KMA events within 0.1 s (about 0.8 km) at every station with both picks.
- **Waveforms.** At the stations where they have signal, 19 correlate with the M_L 2.2 above the noise baseline (0.46 for P, 0.41 for S). 15 reach the relocation threshold of 0.7.

No. 11 (2021-08-27 02:09) and No. 12 (2021-08-27 18:44) fail: the first comes from elsewhere (a shorter S−P interval, a location a few kilometers away and deeper); the second is too weak to decide. Template matching and the relocation decide such cases with more stations.

Size. Relative magnitudes come from S-wave amplitude ratios (5–20 Hz) against the KMA events of the same year. Left out one at a time, the KMA events are reproduced with an RMS difference of 0.30 (0.37 in 2021, 0.20 in 2026). The new events are −0.6 to 1.1 (median 0.3), below the KMA events (Figure 11). The 2021 values rest on about 6 stations each and may be low by up to 0.60.

Timing (Figure 12). 3 new events (18 July 14:50, 19 July 02:56, 19 July 03:38 UTC) preceded the first KMA-listed event of 2021 by 1.5–2.1 days. Their S−P intervals match within 0.02 s, and their S waves correlate with the M_L 2.2 at 0.83–0.88. 3 more (09:54, 10:00, 10:10 UTC on 26 August) preceded the 27 August events, and 3 (5, 9, 16 February) followed the last KMA-listed event of the 2026 episode. The KMA-listed event of 26 May 2026 lies on the patch after the second episode. The pilot covered only parts of the two years; production detection over the whole window (G4) and template matching (G7) will show whether activity started earlier and how often the patch was active between the episodes (Q1).

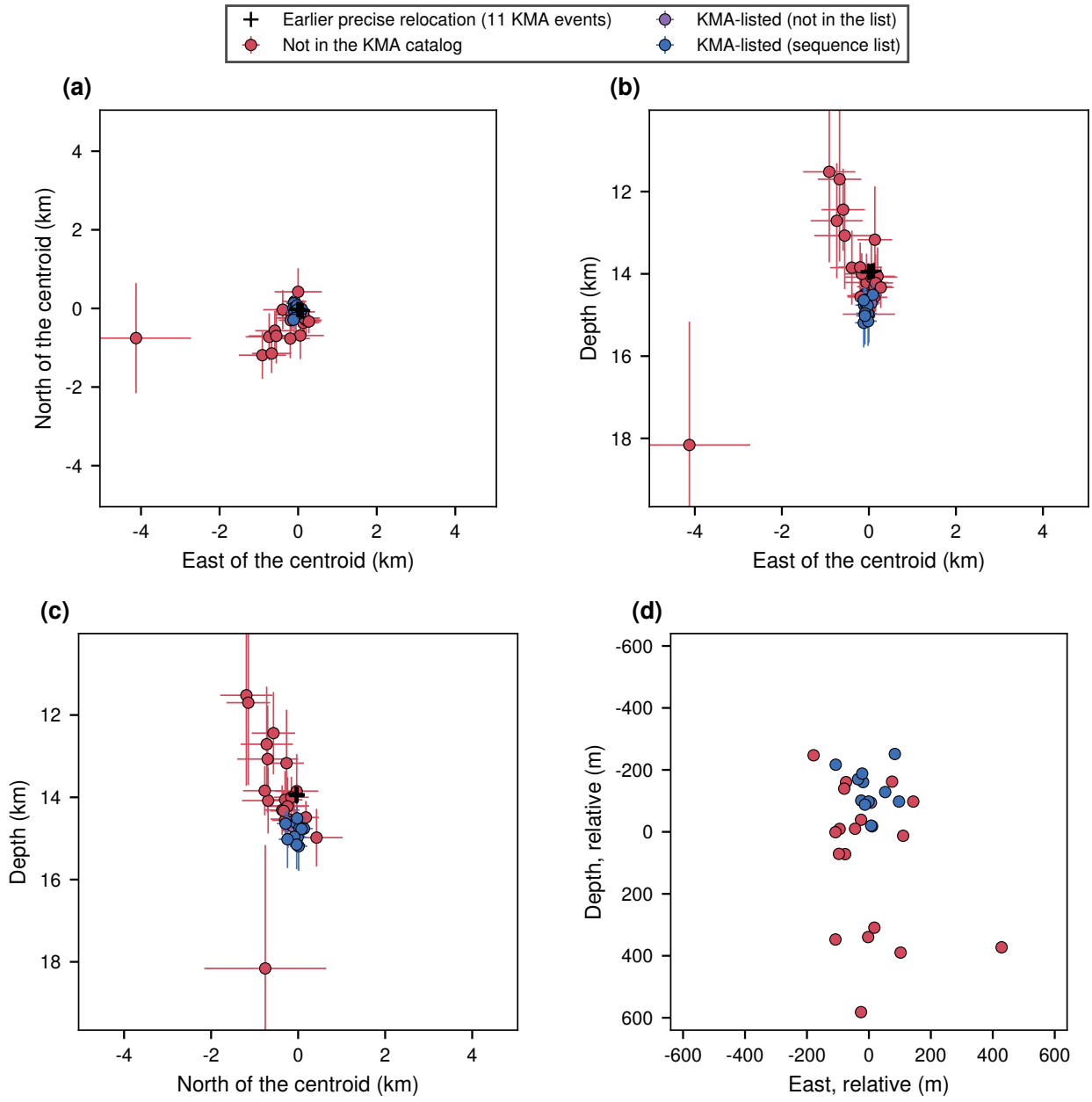


Figure 7. Pilot events, relative view. (a) Map view and (b, c) depth sections of the HYPOINVERSE locations in km from the cluster centroid, with HYPOINVERSE errors (ERH horizontally, ERZ vertically). Black crosses: the earlier precise relocation of the 11 sequence events. (d) Double-difference relocation from the picks alone, relative to its mean.

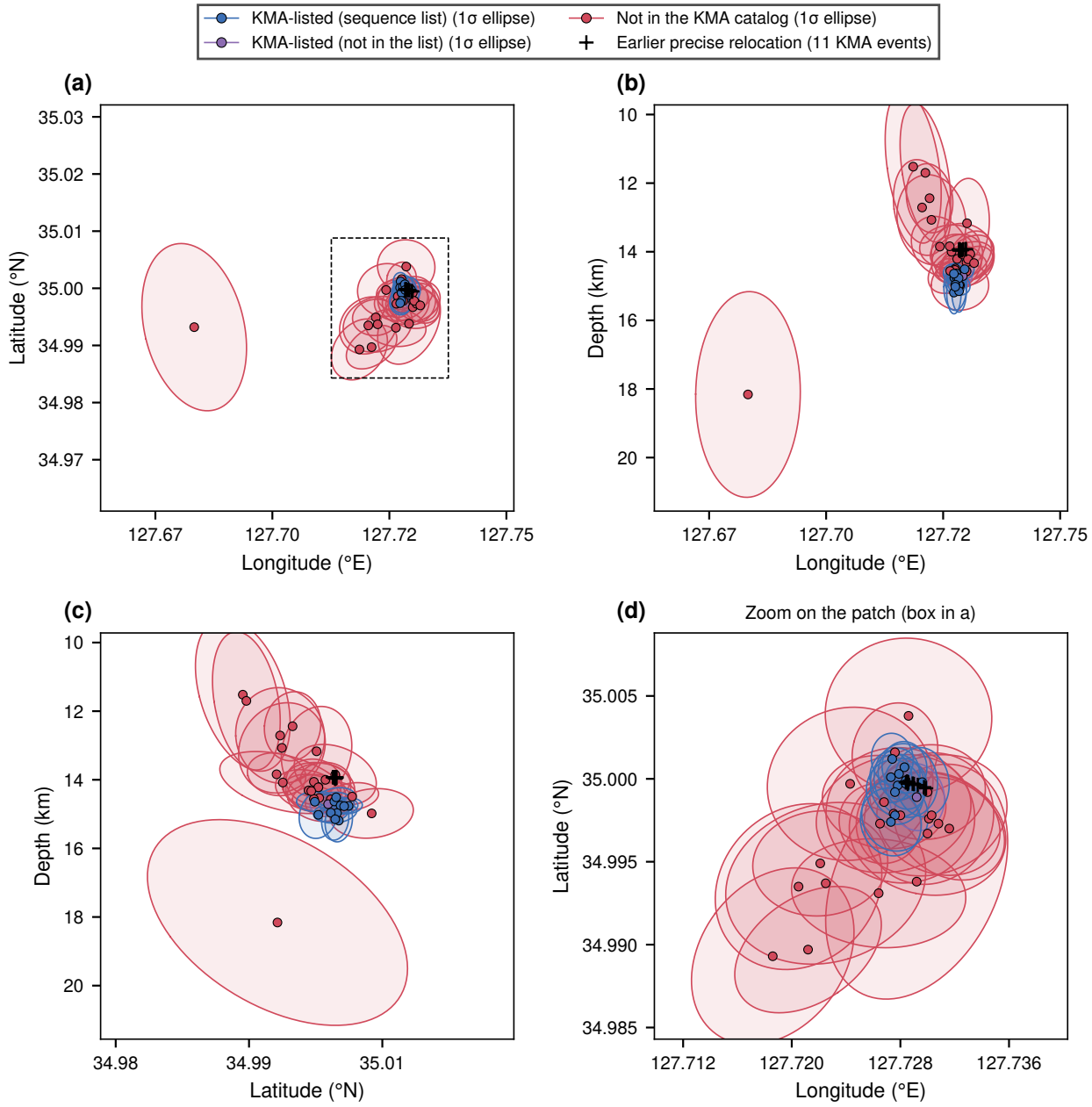


Figure 8. Pilot events, absolute view: HYPOINVERSE locations with 1σ error ellipses (the three principal error axes of each event, from HYPOINVERSE's archive file, projected onto each plane). Black crosses: the earlier precise relocation. (d) Zoom on the patch (box in a).

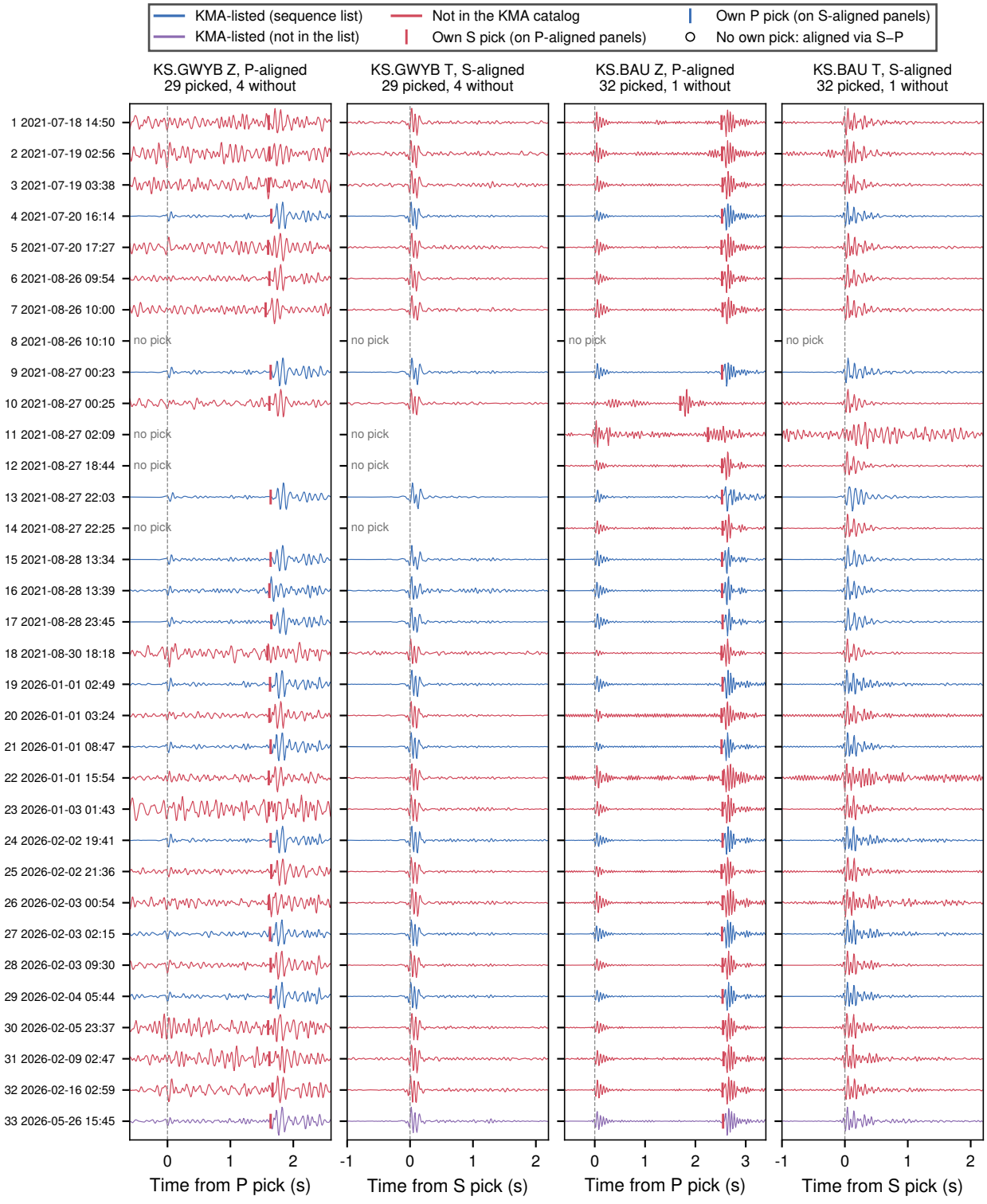


Figure 9. Waveforms of the 33 pilot events at KS.GWYB (7 km) and KS.BAU (17 km), 5–20 Hz, normalized. Vertical components are aligned on each event's own P pick, transverse components on its own S pick; ticks mark the other pick, so that equal S−P intervals line up. Events without a pick at a station are labeled.

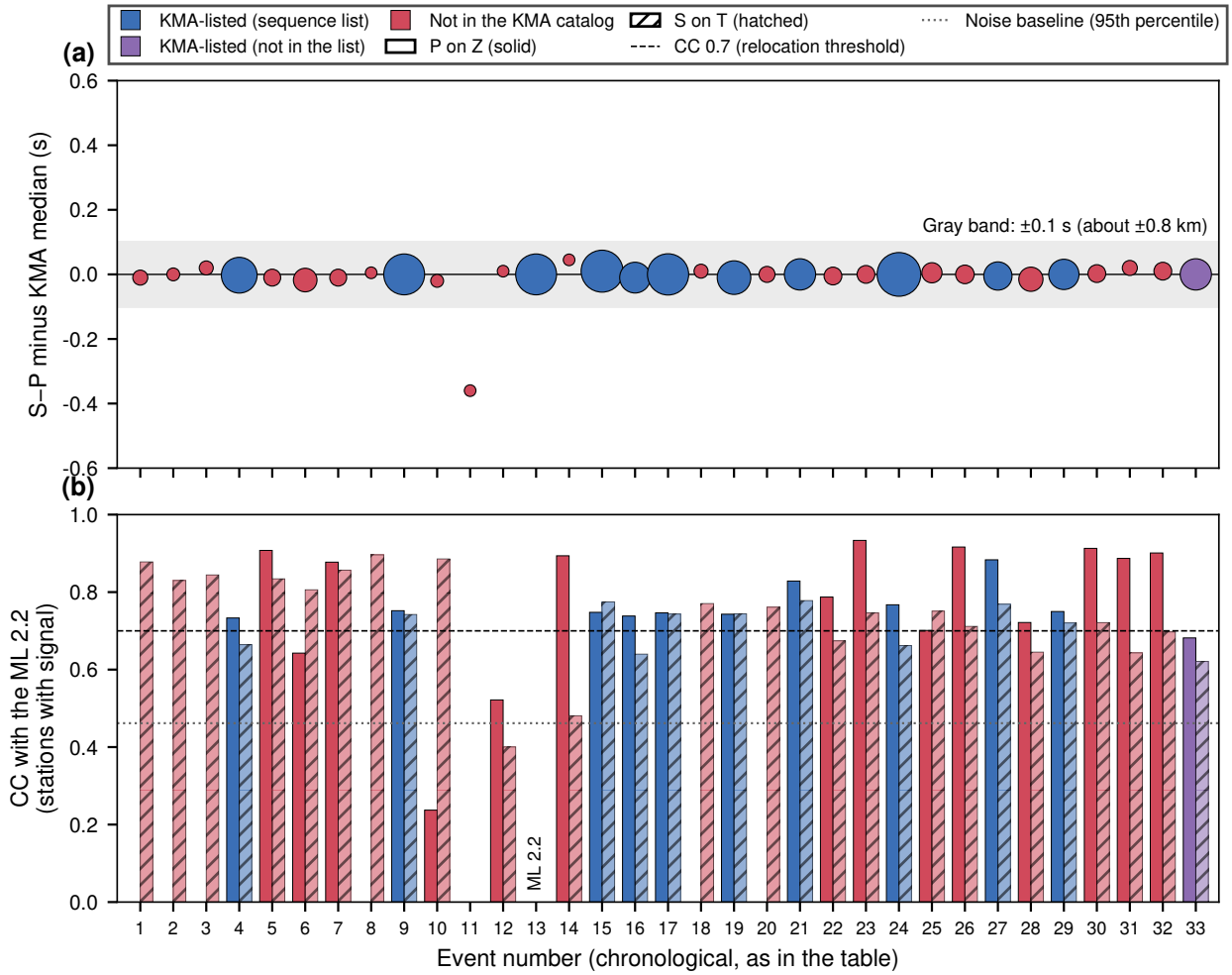


Figure 10. Are the pilot events on the same patch? (a) Each event's S−P interval minus that of the KMA events at the same station (median over the stations with both picks; symbol size grows with the number of stations); gray band ±0.1 s. (b) Cross-correlation with the M_L 2.2 (5–20 Hz) at the stations where the event has signal, median over those stations; P on the vertical solid, S on the transverse hatched. Dashed: the relocation threshold; dotted: the noise baseline.

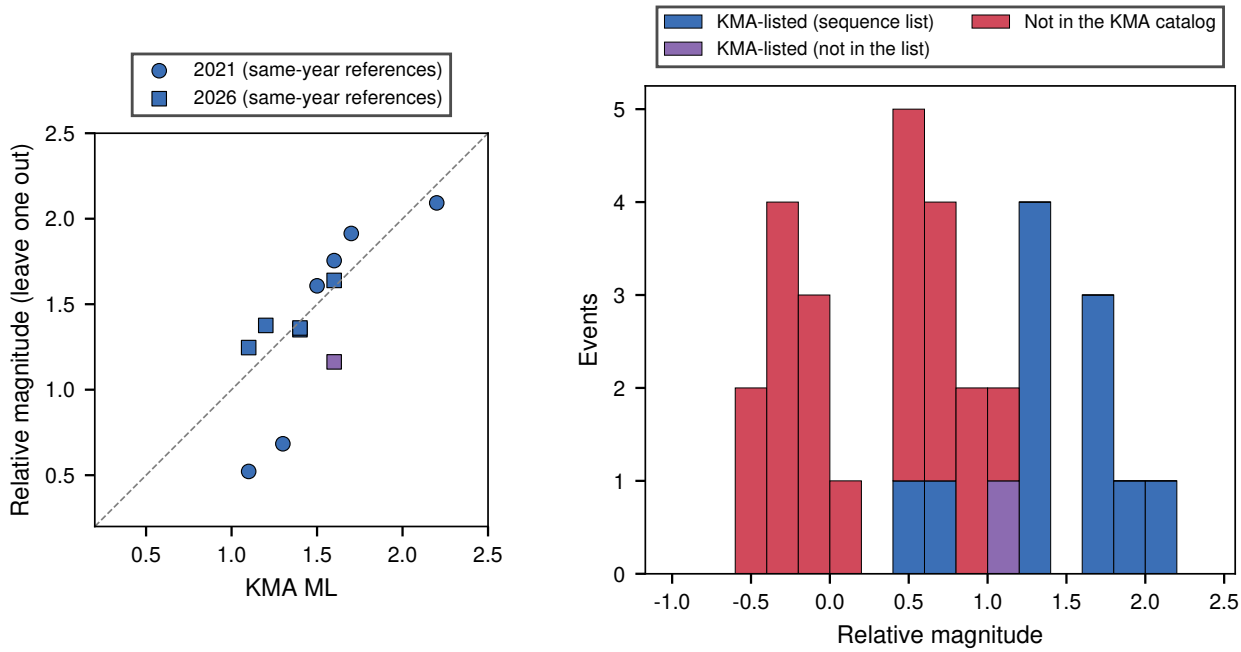


Figure 11. Relative magnitudes from S-wave amplitude ratios. (a) Each KMA-listed event estimated from the other KMA-listed events of the same year, against its KMA M_L. (b) Distribution for all 33 pilot events.

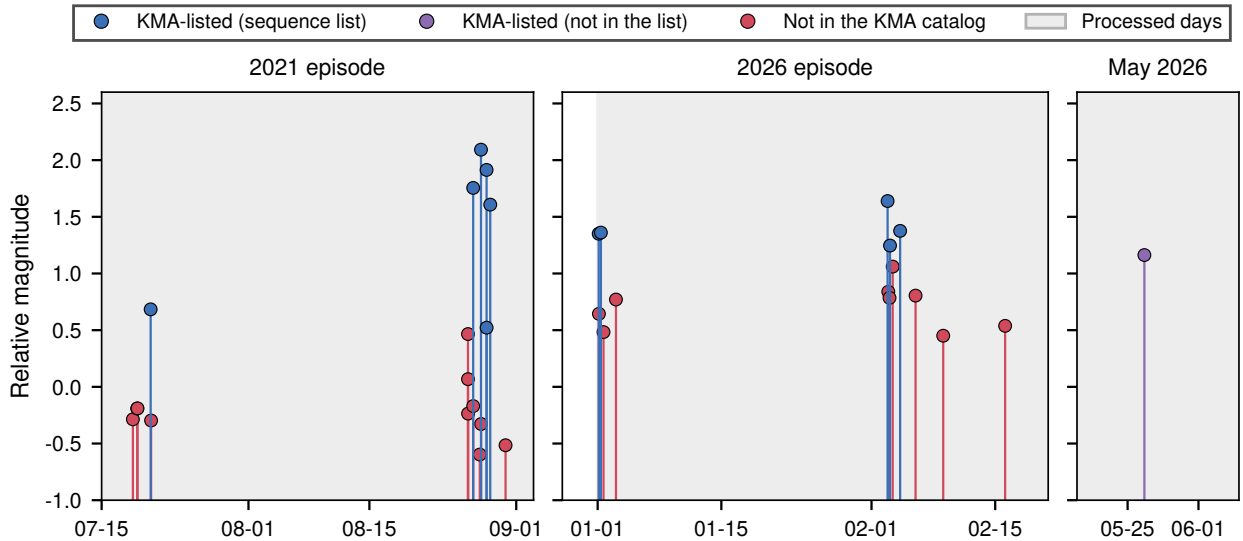


Figure 12. The pilot events in time, each episode on its own axis; processed days shaded.

8 Independent checks

A check counts only if it does not reuse the quantity being fitted. The study uses:

1. recovery of the KMA-listed events: the 14 events within 5 km (the 11 of the sequence list and three more, decision R5), the other events within 10 km, and the regional catalog;
2. the earlier relocation: after one rigid shift, the relative positions of the 11 sequence events should agree within their uncertainties;
3. S-wave times on the transverse component against those on north and east, at stations whose orientation never changed: they should agree to about 1 ms;
4. GWYB templates of 2021 found again on days in each of GWYB's four orientation periods, which tests the turning of the templates;
5. every template detecting its own event;
6. the local versus the national orientation measurements (done: median difference 0.00°);
7. the smoke test of the detection chain (done, Table 1).

9 Status and schedule

At 2026-10-16 16:40 KST, stages G0–G3 are complete.

- **Downloads.** MARA: all 760 station-sensor-years, 6.9 TB, no problems. KIGAM: 10,629 files, 0 errors. NECIS gap filling done.
- **Inputs.** The link tree holds 711,715 links. The final orientation run and the final sensor choice are done.
- **Freeze.** The detection parameters were frozen at 2026-10-16 12:51 (hash d724acfddcd608dd).

Production detection (G4) needs moho's graphics processor. The Haenam template-matching scan holds it; the Haenam work has priority, and the scan should end around 19 October. Then, in order:

1. a short run of the pilot relocation on the graphics processor, to check the cross-correlation stage on the 33 pilot events;
2. production detection over all years: about two days with four chains, at the Haenam rate of about 2.75 s per station-day per chain;
3. relocation (G5) and magnitudes (G6);
4. template matching (G7), timed on the episode days first; about five days at the Haenam rate of about 1.5 min per day;
5. the joint relocation (G8) and the analysis (G9).

10 Risks and limitations

- **A changing network.** With 8 stations within 60 km in 2013 against 24 in 2021, detection is less sensitive in the early years. Activity before 2016 can only be compared with later activity above an era-specific completeness magnitude (Q5).
- **Links across instrument changes.** Cross-correlation needs similar waveforms. A sensor replaced between two events lowers the correlation, so early events may be tied to the later ones by fewer measurements. This is reported per era.
- **Orientation gaps.** On days inside a change bracket, or at a sensor never measured, S is not measured on the transverse component and the template's S windows get no weight. After the final orientation run, 6 chosen sensors remain unmeasured, all short-lived accelerometers of 2016–2018.
- **Shared graphics processor.** Gwangyang's GPU stages run only when the GPU is free of the higher-priority Haenam work, so the schedule follows Haenam's.

- **Data holes.** The vertical components still missing on the M_L 2.2 days (BOSB, GMNA) and the HWJA outage of January–May 2026 cannot be filled.
- **False detections over 14 years.** Long template-matching scans pick up noise bursts and events seen by a single template. Each candidate group is vetted in its own notebook, as for Imsil.
- **Open decisions:** G-D12, G-D19 (Appendix A). G-D12 is the use of the early 20-sample-per-second channels; G-D19 is the list of templates excluded from the catalog, which is re-derived when the templates exist.

11 Deliverables

- catalogs: the detection catalog 2012–2026; the relocated catalog with its 25 m reference set and fault plane; local magnitudes; the template-matching catalog; the joint relocation of the detections;
- stage notebooks 00–10, each with figures, checks and a computed summary, and the decision log;
- a results report built like this one, in which every number comes from a notebook;
- on request, a public code repository with server names and credentials removed, as for Imsil.

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A Decision log

All settings of this study as recorded in `docs/decisions.csv` (34 rows, 20 of them Gwangyang decisions G-D). P: decided before the work started; R: region values set by rule; G-D: settings this study had to choose, with the evidence in the log. Status *default* means the plan's default, applied because the stages run unattended; such a setting can be changed on review.

ID	Setting	Gwangyang value	Status
P1	study window	2012-12-20 (first KS.GWYB day) .. 2026-10-06 (last day on MARA, 2026-10-15 listing)	decided
P2	machine and project folder	moho /home/msseo/works/73.Gwangyang_sequence	decided
P3	method rules	the same: one sensor per station, one dated orientation table, S dt.cc on T, template N/E turned to each scanned day, ML without station terms	decided
P4	networks	KS + KG where data exist; JU not used	decided
R1	association box	34.30-35.70 N, 126.93-128.53 E	rule
R2	source centroid	35.0000 N 127.7286 E	rule
R3	relocation selection	within 5 km of 35.0000 N 127.7286 E, depth ≥ 10 km, same quality gate (relocation_selection.py)	rule
R4	station set	115 KS/KG codes within 100 km with data in the window (107 KS, 8 KG; 5 of them temporary codes of 2023; KS.KWJ excluded, G-D20)	rule
R5	known events	registry/events.csv: the 11 KMA events of the two episodes (UTC); all 14 KMA events within 5 km in the window (NECIS catalog to 2026-10-13) are the G4 recovery check	rule
G-D1	sensor per station	the sensor rule for every multi-sensor station (after the final G2 run of 2026-10-16 it applies everywhere; GWYBT and NAWBT resolved); no choice changed by G2: HG 85, HH 16, EL 14 (temporary codes included)	default
G-D1b	sensor rule over a 14-year window: coverage first	a sensor whose data period is > 365 d shorter than the station's longest sensor is not eligible; the discontinuity rule decides among the rest	default
G-D2	unmeasured orientation spans	the same; G2 measures the gaps it can (GWYB 2012-12..2013-01, KS HG 2016-2017, after the national table ends)	default
G-D3	relocation depth floor	10 km	default
G-D4	template-matching station subset	within 60 km (the plan default), one sensor each; temporary codes read as their site; template_matching/stations_G7_60km.csv	default
G-D5	mirror bandwidth	8 jobs, rsync bwlimit 25 MB/s each, one ssh connection per job; measured total about 59 MB/s (the MARA-moho link limit)	default
G-D6	template-matching threshold	10 x MAD per configuration segment, no floor (Imsil D23); compared on the 2021 and 2026 days before the long scan	default
G-D7	temporary codes (2023: GWYBT, NAWBT, SUCAT, DAYAT, GEJAT)	merged: linked only on days the site's chosen sensor lacks	default
G-D8	KG after 2024-10-31	superseded by G-D14 (KIGAM FDSN fill of the chosen KG sensors)	superseded
G-D10	predecessor codes at one site	older code linked until the day before its successor starts (11 codes); KWJU and KWJ2 both used (renewal decision B)	default
G-D11	KG stations also filed under KS on MARA (2017)	KG used; the KS copies (GRE, GSU, HDB, JRB, MUN, YSB) ignored	rule
G-D12	20 Hz SL channels (YSU, TOY, KCH, MAS, JEU before 2015-04; GRE, KJM)	not used: the frozen PhaseNet+ runs at 100 Hz and the 5-20 Hz cross-correlation band exceeds SL's 10 Hz Nyquist	open
G-D13	station-days with a missing component on MARA	audit every chosen station-day after the pull; re-fetch from NECIS for the 2021 and 2026 episode days (KMA source archive gaps); elsewhere the components present are used, as the frozen pipeline does	default

ID	Setting	Gwangyang value	Status
G-D14	KG data after 2024-10-31	the same tool, only the chosen sensor of each KG station (station table Sensor): JRB, WDL, GSU (HG) and GRE, KJM (EL) 2024-11-01..2026-10-06; HWSB, GHR, IBA (HG) only 2026-10-01..10-06, as MARA holds them through 2026-09-30 (the Haenam R1a fill delivered to MARA)	default
G-D15	template-matching scan settings other than the threshold	the same; period 2012-12-20..2026-10-06; templates = focused 25 m set of the G5 run (catalogs/templates/build_template_catalog.py); picks from that run only	default
G-D16	25 m reference adoption	the same definitions anchored on the 2021 ML 2.2 (2021-08-27 22:03:22 UTC); eras before 2021 / 2021 episode (Jul-Sep) / interval / 2026 episode (Jan-Feb) / after	default
G-D17	NAHA in the local magnitudes	NAHA kept: the station table carries its corrected site (34.8167 N, 127.9261 E), and the SAC exports take station coordinates from the station table	default
G-D18	joint relocation of the template-matching detections (G8)	Imsil D22 with S on T (R/T exported at each detection origin), each detection exported with the station table of its own year, and the G5 onset window for the 2021 ML 2.2 when it is a reference	default
G-D19	templates excluded from the template-matching catalog	re-derived at G7 with the same test on the Gwangyang templates; empty until then	open
G-D20	station codes excluded for unusable data	KS.KWJ excluded from the inventory and station table	default
G1	data (result)	MARA pull 717,147 files (6,274.6 GB) + coverage-first supplement 64,336 files (620.7 GB), 0 problems; verify: only 2026-10-08 missing (after the window); KIGAM 10,629 files, 0 errors; NECIS 9 of 11 gap requests recovered; link tree 711,715 links, check PASS	result
G2	orientation (result)	22,078 station-teleseism windows measured from local day files; 147 P_minT periods for 110 sensors (144 valid); local vs national: 950 events accepted by both, |diff| median 0.00, 95% 0.10, max 0.30 deg; 6 chosen sensors unmeasured (YES, SAC, HAC, JAS, JAH, CHR: accelerometers of 2016-2018)	result
G3	smoke test (2021-08-27/28, 2026-01-01, 2013-07-19)	PASS: 7/7 KMA events within 1.23 s; ML 2.2 0.12 km from the earlier relocation (depth 15.2 km HYPOINVERSE); PhaseNet+ - manual median |dt| P 0.019 s, S 0.020 s (n 458)	result
G4-freeze	detection parameter freeze	hash d724acfddcd608dd (config, core, pick augmentation, station table of 115 codes, PhaseNet+ weights sha f1643330...), 2026-10-16 12:51; smoke-test outputs moved to detection/smoke_G3_20261015	result
G6-resp	merged response inventory (result)	docs/response_inventory/merged_inventory.xml: 636 channel epochs in 115 stations; 692,785 of 706,261 channel-days covered (98.1 percent); both episodes without a day lacking a response (2021: 61 streams, 2026: 82); 1,770 channel-days without response (station starts before their metadata in early 2018, YSU 2016, SCYA 2025-12, HAMA 2024-01..03); KWJU excluded (neither source fits its data, as in Haenam); remove_response spot checks at 5 channels give ratios 0.998-1.001	result

B Pilot events

The 33 events of the pilot (notebook 91). Origin, location, phases used (Ph.), azimuthal gap and errors from HYPOINVERSE. Δ(S−P): median difference between the event's S−P intervals and those of the KMA events at the same stations. CC_S: median correlation of the S wave with the M_L 2.2 at the stations where the event has signal. Rel. M: relative magnitude. Class: KMA (sequence list), KMA* (KMA-listed, not on the list), new.

No.	Origin (UTC)	Lat. (°N)	Lon. (°E)	Depth (km)	Ph.	Gap (°)	ERH (km)	ERZ (km)	Δ(S−P) (s)	CC_S	Rel. M	KMA ML	Class
1	2021-07-18 14:50:01	34.9978	127.7280	14.21	15	86	0.5	0.7	−0.01	0.88	−0.3		new
2	2021-07-19 02:56:11	34.9935	127.7205	12.71	9	111	0.6	1.4	0.00	0.83	−0.2		new

No.	Origin (UTC)	Lat. (°N)	Lon. (°E)	Depth (km)	Ph.	Gap (°)	ERH (km)	ERZ (km)	Δ(S−P) (s)	CC_S	Rel. M	KMA ML	Class
3	2021-07-19 03:38:12	34.9997	127.7243	13.85	12	109	0.5	0.9	+0.02	0.84	−0.2		new
4	2021-07-20 16:14:18	34.9996	127.7278	14.63	86	35	0.2	0.3	0.00	0.66	0.7	1.3	KMA
5	2021-07-20 17:27:05	34.9949	127.7221	12.44	16	103	0.5	1.0	−0.01	0.83	−0.3		new
6	2021-08-26 09:54:43	35.0016	127.7276	14.49	43	40	0.3	0.4	−0.02	0.81	0.5		new
7	2021-08-26 10:00:25	34.9976	127.7301	13.17	19	86	0.4	1.3	−0.01	0.86	0.1		new
8	2021-08-26 10:10:18	35.0038	127.7286	14.98	10	86	0.6	0.7	+0.01	0.90	−0.2		new
9	2021-08-27 00:23:15	35.0012	127.7274	14.76	114	35	0.2	0.2	0.00	0.74	1.8	1.6	KMA
10	2021-08-27 00:25:42	34.9938	127.7292	14.08	12	112	0.6	0.8	−0.02	0.89	−0.2		new
11	2021-08-27 02:09:29	34.9932	127.6833	18.16	8	140	1.4	3.0	−0.36				new
12	2021-08-27 18:44:02	34.9893	127.7186	11.52	8	113	0.6	2.2	+0.01	0.40	−0.6		new
13	2021-08-27 22:03:23	35.0001	127.7273	15.19	114	35	0.2	0.6	0.00		2.1	2.2	KMA
14	2021-08-27 22:25:31	34.9897	127.7212	11.70	10	108	0.5	2.0	+0.04	0.48	−0.3		new
15	2021-08-28 13:34:11	35.0003	127.7279	14.75	115	35	0.2	0.2	+0.01	0.77	1.9	1.7	KMA
16	2021-08-28 13:39:22	34.9999	127.7284	14.97	71	39	0.2	0.3	−0.01	0.64	0.5	1.1	KMA
17	2021-08-28 23:45:56	35.0007	127.7283	14.77	114	35	0.2	0.2	0.00	0.74	1.6	1.5	KMA
18	2021-08-30 18:18:23	34.9937	127.7225	13.07	9	111	0.7	1.3	+0.01	0.77	−0.5		new
19	2026-01-01 02:49:47	34.9992	127.7276	14.96	82	40	0.2	0.3	−0.01	0.74	1.4	1.4	KMA
20	2026-01-01 03:24:59	34.9967	127.7300	14.31	17	70	0.4	0.6	0.00	0.76	0.6		new
21	2026-01-01 08:47:59	34.9998	127.7296	14.51	78	40	0.2	0.3	0.00	0.78	1.4	1.4	KMA
22	2026-01-01 15:54:00	34.9973	127.7308	14.06	23	41	0.4	0.7	0.00	0.67	0.5		new
23	2026-01-03 01:43:12	34.9979	127.7275	14.53	20	70	0.4	0.7	0.00	0.75	0.8		new
24	2026-02-02 19:41:19	34.9997	127.7284	15.15	127	35	0.2	0.6	0.00	0.66	1.6	1.6	KMA
25	2026-02-02 21:36:20	34.9992	127.7300	14.57	30	54	0.3	0.5	+0.01	0.75	0.8		new
26	2026-02-03 00:54:43	34.9986	127.7268	14.00	26	50	0.4	0.5	0.00	0.71	0.8		new
27	2026-02-03 02:15:35	34.9978	127.7276	15.02	62	40	0.2	0.7	−0.01	0.77	1.2	1.1	KMA
28	2026-02-03 09:30:50	34.9973	127.7265	14.56	47	41	0.3	0.4	−0.01	0.65	1.1		new
29	2026-02-04 05:44:06	34.9974	127.7273	14.64	67	36	0.2	0.3	0.00	0.72	1.4	1.2	KMA
30	2026-02-05 23:37:57	34.9978	127.7303	14.22	19	86	0.4	0.7	0.00	0.72	0.8		new
31	2026-02-09 02:47:34	34.9931	127.7264	13.84	15	104	0.5	0.6	+0.02	0.64	0.5		new
32	2026-02-16 02:59:15	34.9970	127.7316	14.33	24	49	0.3	0.5	+0.01	0.70	0.5		new
33	2026-05-26 15:45:44	34.9989	127.7292	14.71	77	35	0.2	0.3	0.00	0.62	1.2	1.6	KMA*